



**Faculty of Engineering & Technology Electrical & Computer  
Engineering Department Circuits and Electronics Laboratory -  
ENEE2103**

**PreLab Exp11: Zener Diodes and Voltage Regulators**

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**Prepared By:**

**Aya Dahbour 1201738**

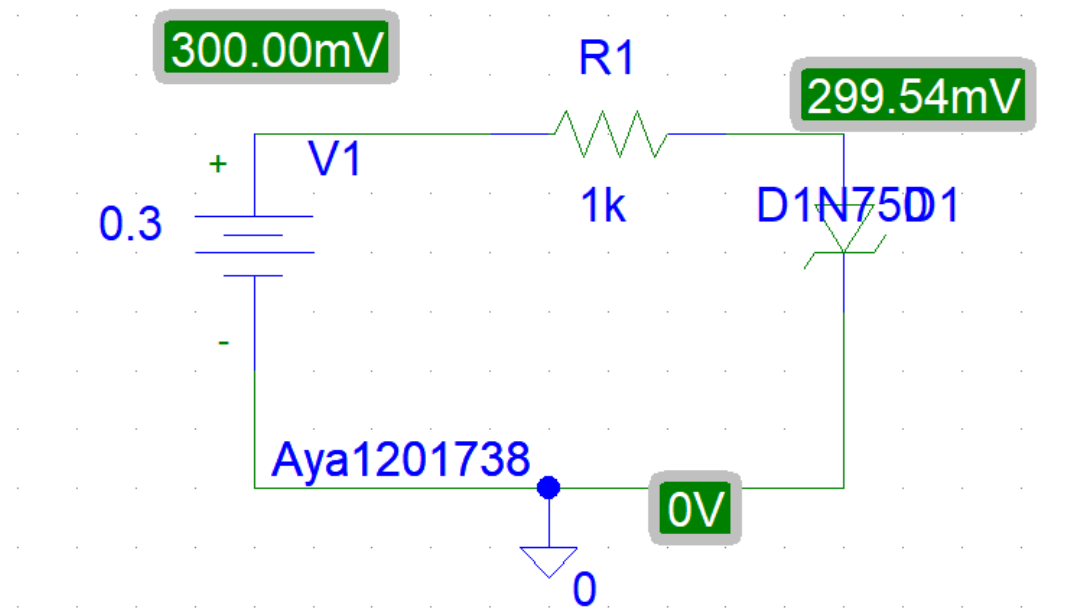
**Instructor: Dr. Nasser Ismail.**

**Assistant: Eng. Bilal Ismail**

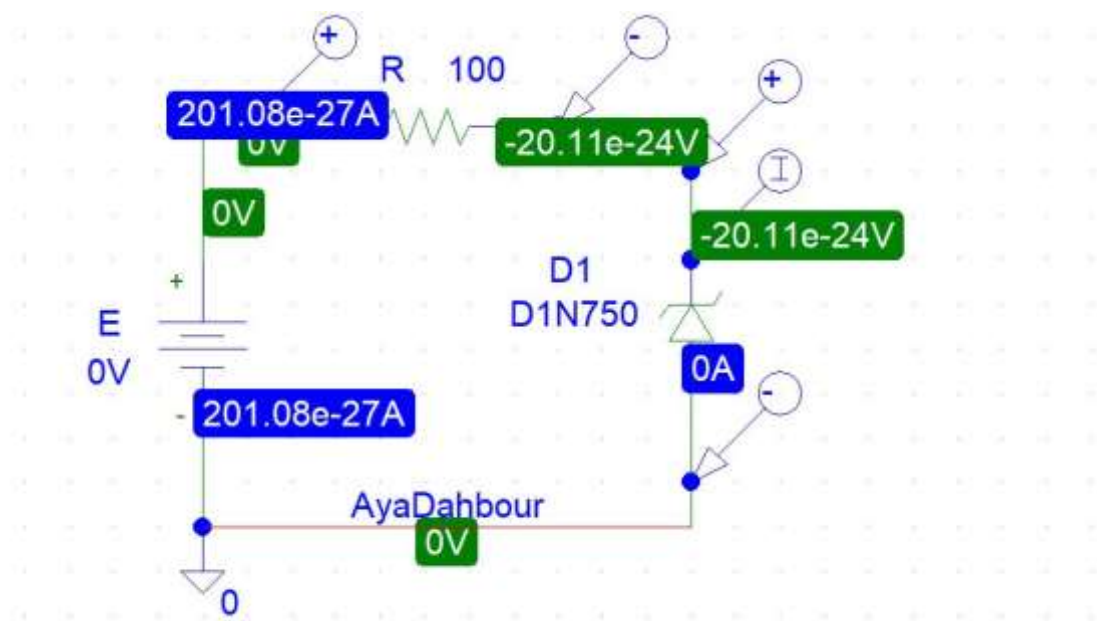
**Section: 3**

**Date: 30/1/2023**

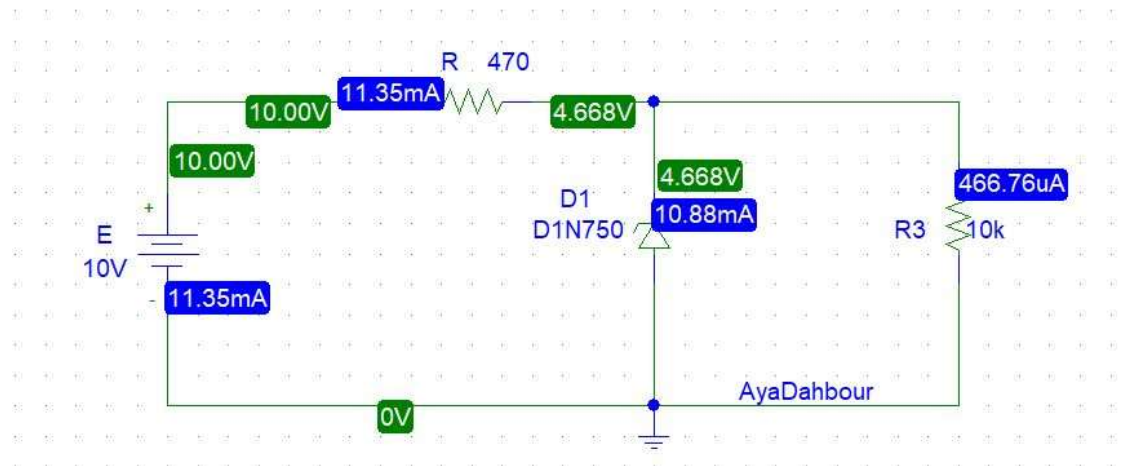
## Zener Diodes:



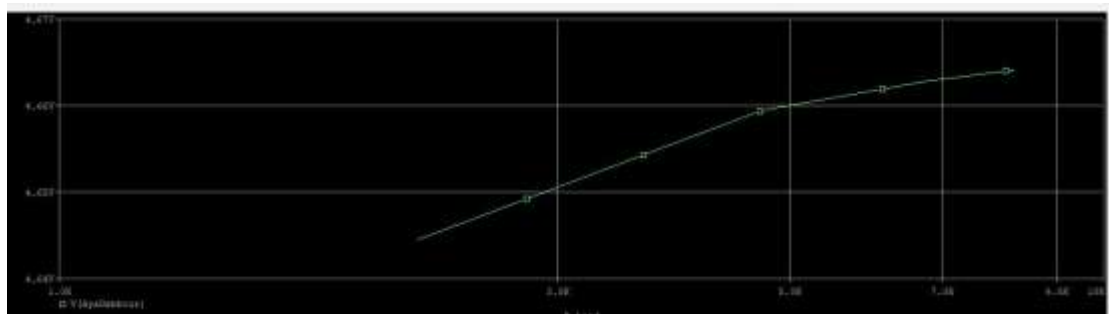
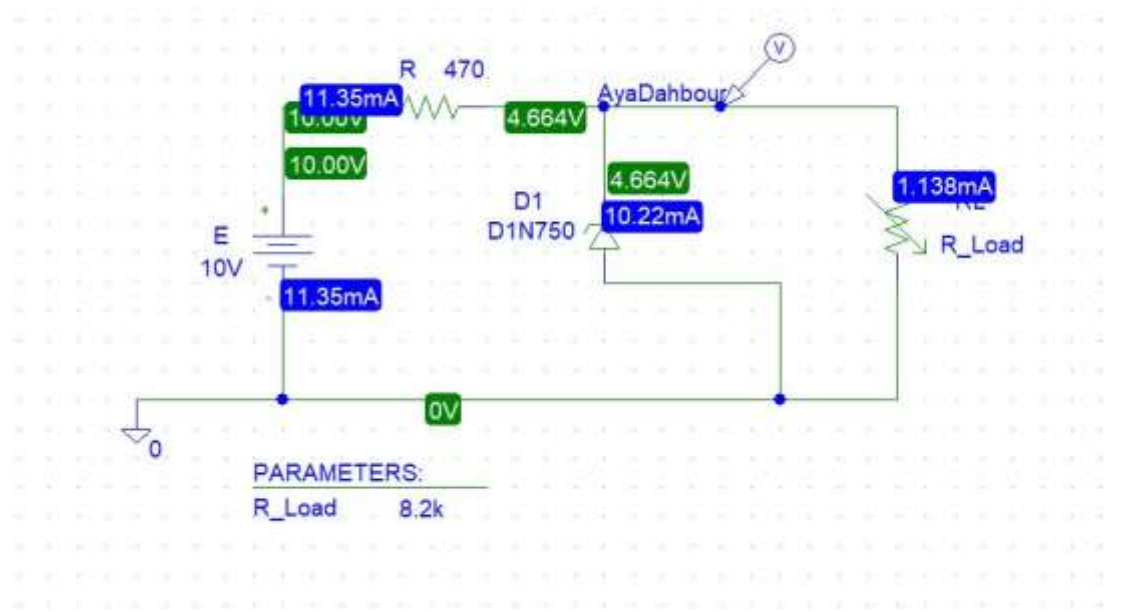
| E(V) | VR (V)   | VZ (V)   | I (mA)   |
|------|----------|----------|----------|
| 0.3  | 0.46mV   | 299.54mV | 495.62nA |
| 0.4  | 2.68mV   | 397.32mV | 2.676uA  |
| 0.5  | 13.05mV  | 486.95mV | 13.05uA  |
| 0.6  | 43.87mV  | 556.13mV | 43.87uA  |
| 0.7  | 98.78mV  | 601.22mV | 98.78uA  |
| 0.8  | 170.35mV | 629.65mV | 170.35uA |
| 0.9  | 251.48mV | 648.52mV | 251.48uA |
| 1    | 338.08mV | 661.92mV | 338.08uA |
| 2    | 1.287V   | 713.19mV | 1.287mA  |
| 3    | 2.268V   | 731.57mV | 2.268mA  |
| 4    | 3.257V   | 742.96mV | 3.257mA  |



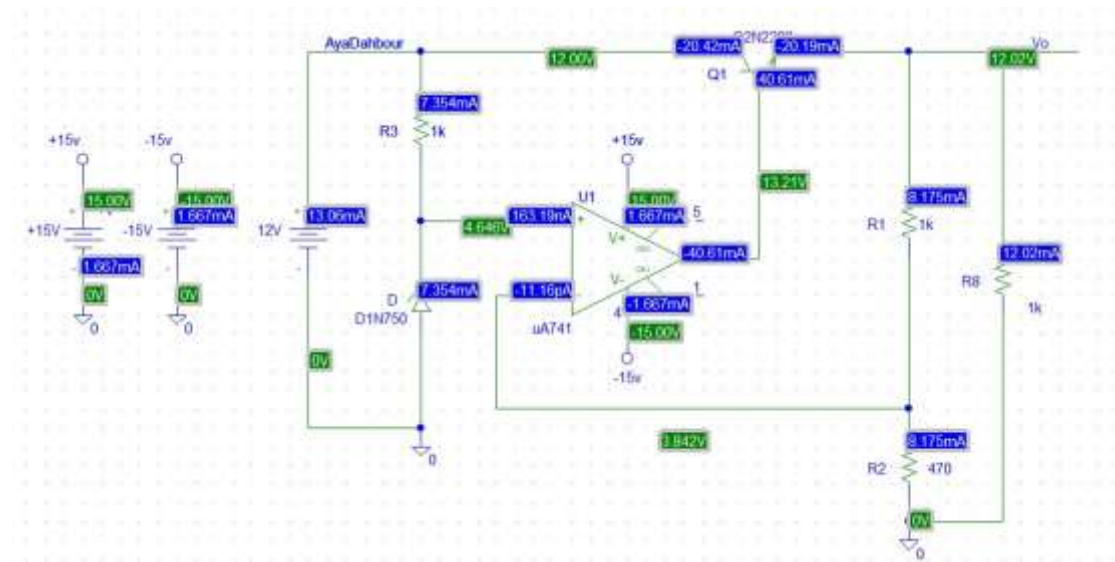
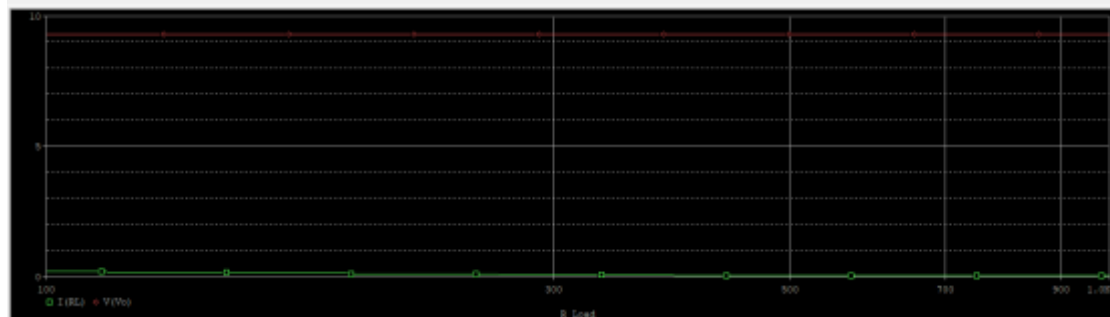
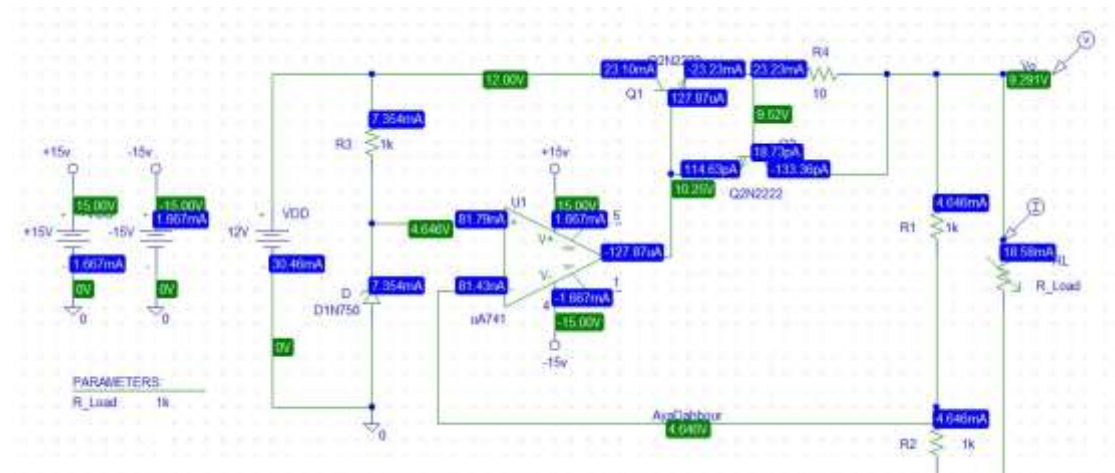
| E (V) | VZ (V) | I (mA)   |
|-------|--------|----------|
| 0.1   | 0.1    | 6.234nA  |
| 0.5   | 0.5    | 40.67nA  |
| 1     | 1      | 141.89nA |
| 2     | 2      | 1.839uA  |
| 3     | 2.998  | 24.13uA  |
| 4     | 3.970  | 297.19uA |
| 5     | 4.607  | 3.928mA  |
| 6     | 4.678  | 13.22mA  |
| 7     | 4.707  | 22.93mA  |
| 8     | 4.727  | 32.74mA  |
| 9     | 4.741  | 42.59mA  |
| 10    | 4.753  | 52.47mA  |
| 11    | 4.763  | 62.37mA  |
| 12    | 4.773  | 72.27mA  |
| 13    | 4.781  | 82.19mA  |
| 14    | 4.789  | 92.12mA  |
| 15    | 7.796  | 102.04mA |

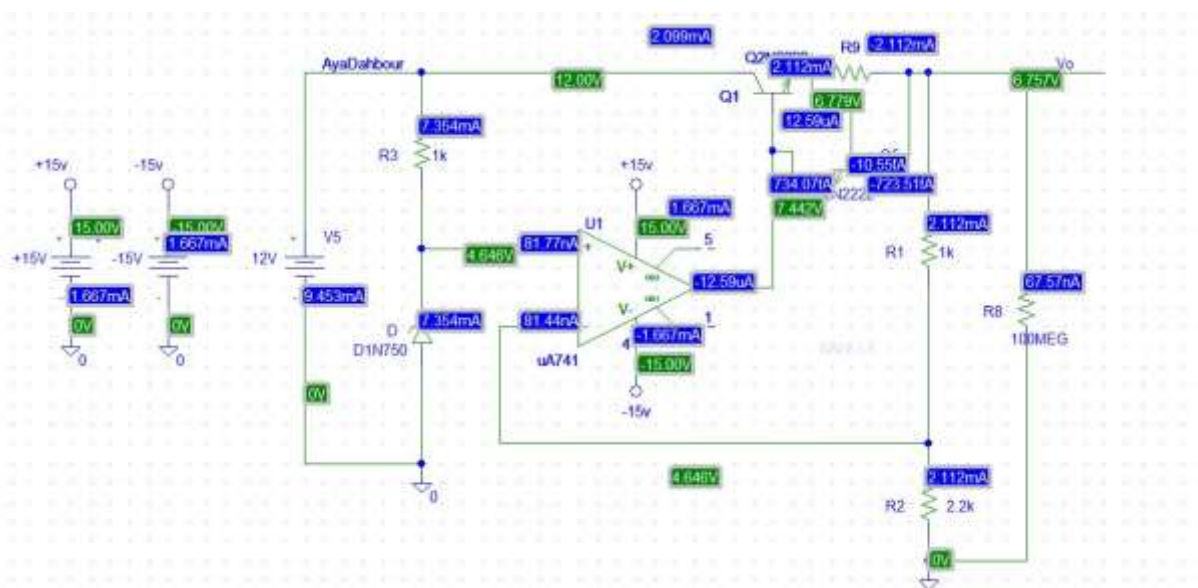
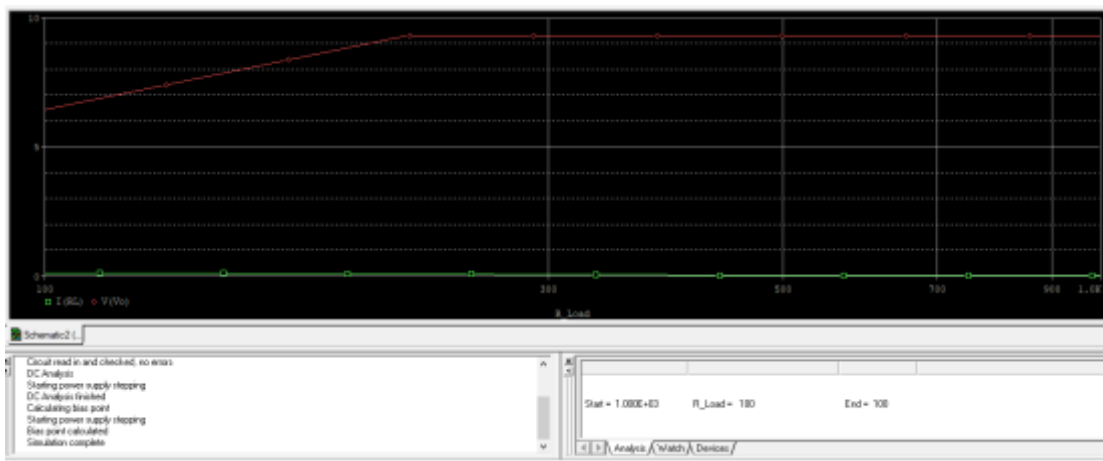
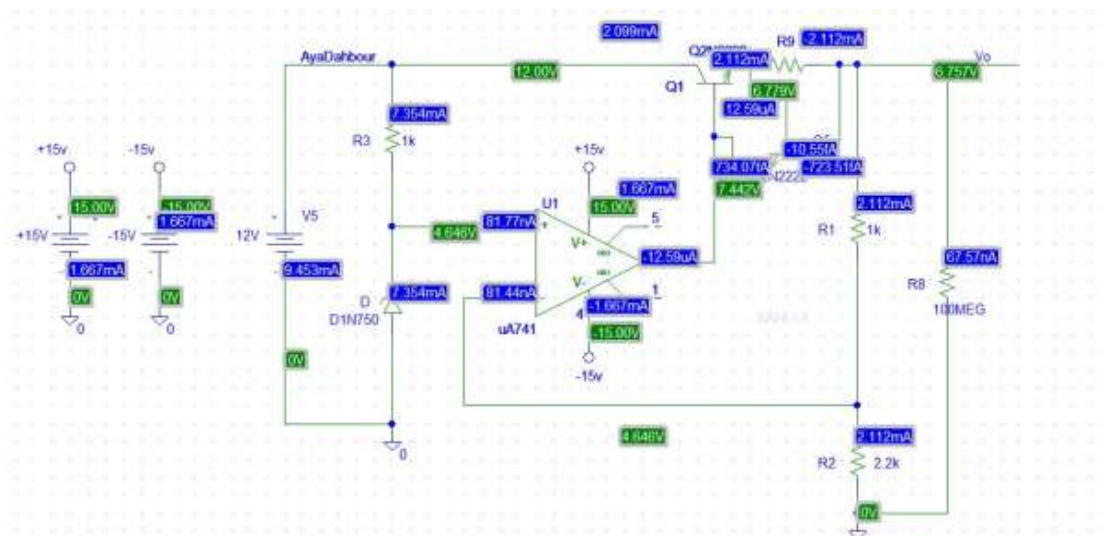


| E     | 10    | 11    | 12    | 13    | 14    |
|-------|-------|-------|-------|-------|-------|
| VL(V) | 4.668 | 4.667 | 4.685 | 4.692 | 4.698 |



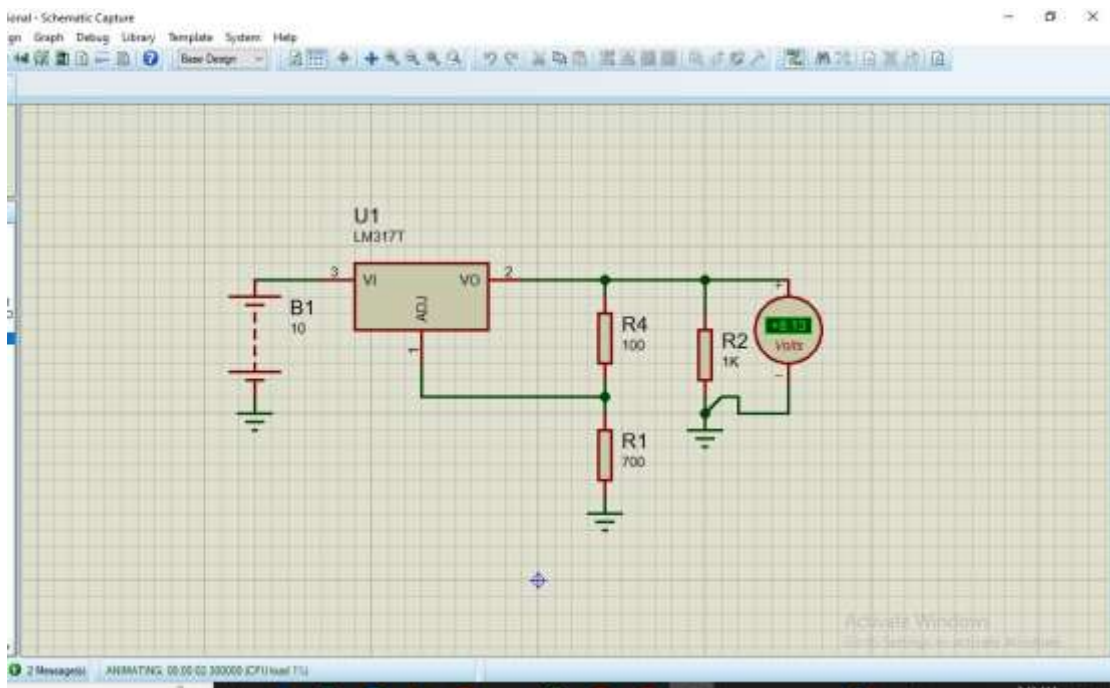
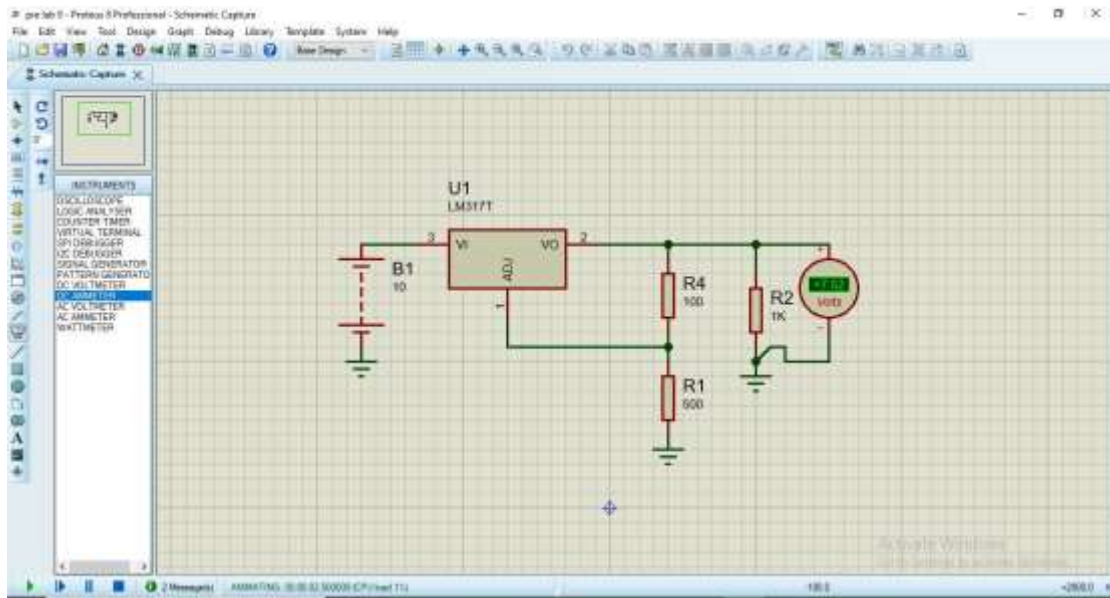
## The Voltage-Regulated Power Supply:



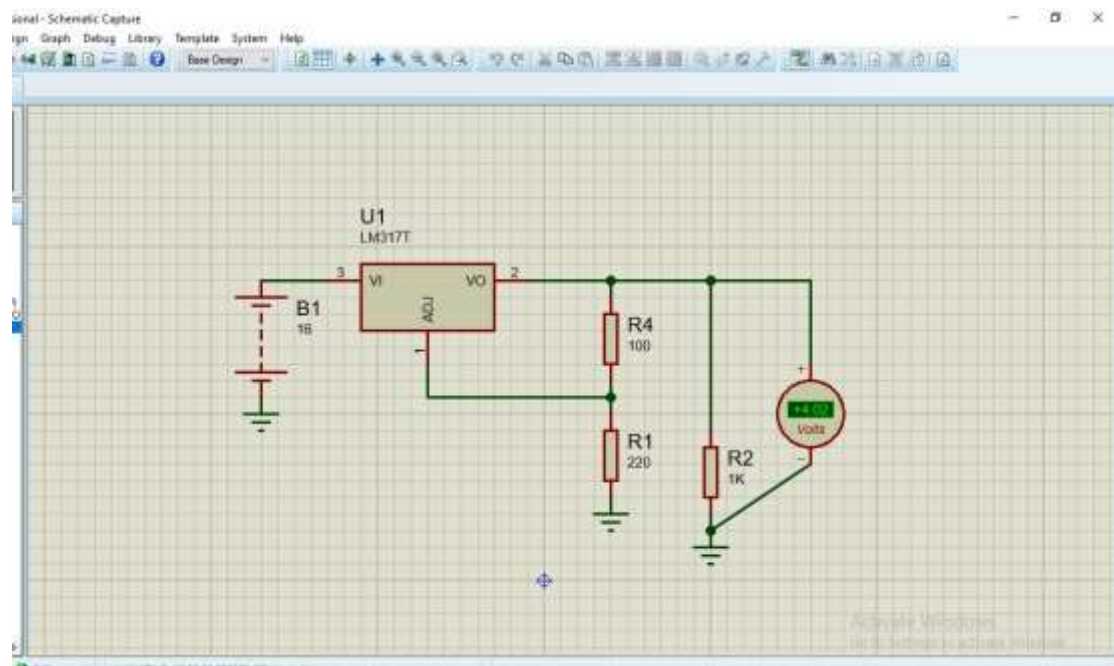
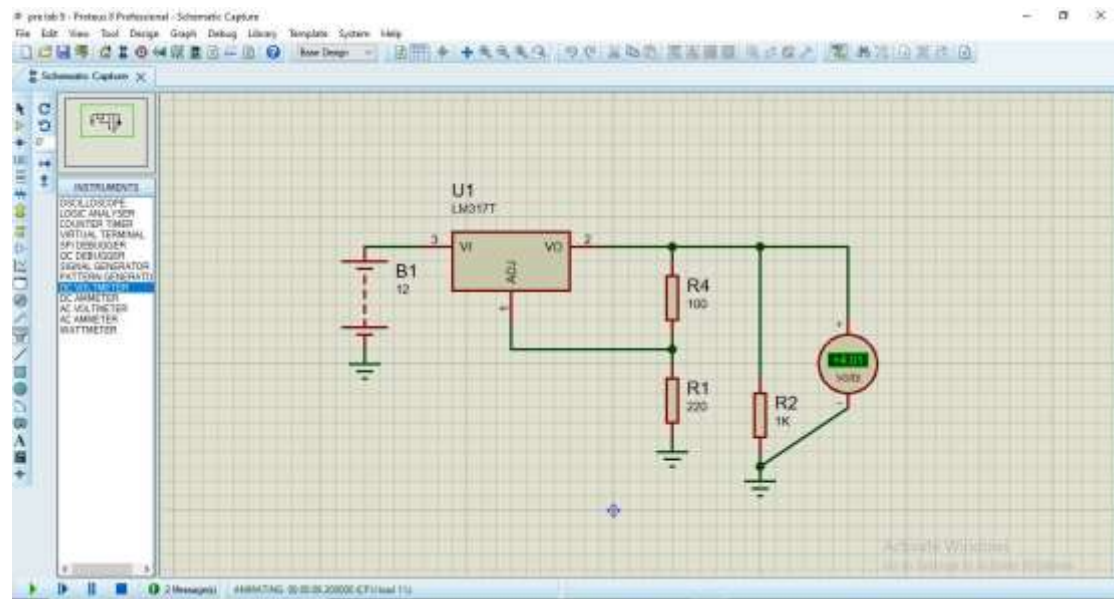


| RL    | open    | 1K      | 680    | 470     | 220     | 100     | 50       |
|-------|---------|---------|--------|---------|---------|---------|----------|
| Vo(V) | 6.757   | 6.757   | 6.757  | 6.757   | 6.757   | 6.757   | 5.666    |
| Io    | 67.57nA | 6.757mA | 9.94mA | 14.38mA | 30.72mA | 67.57mA | 113.32mA |

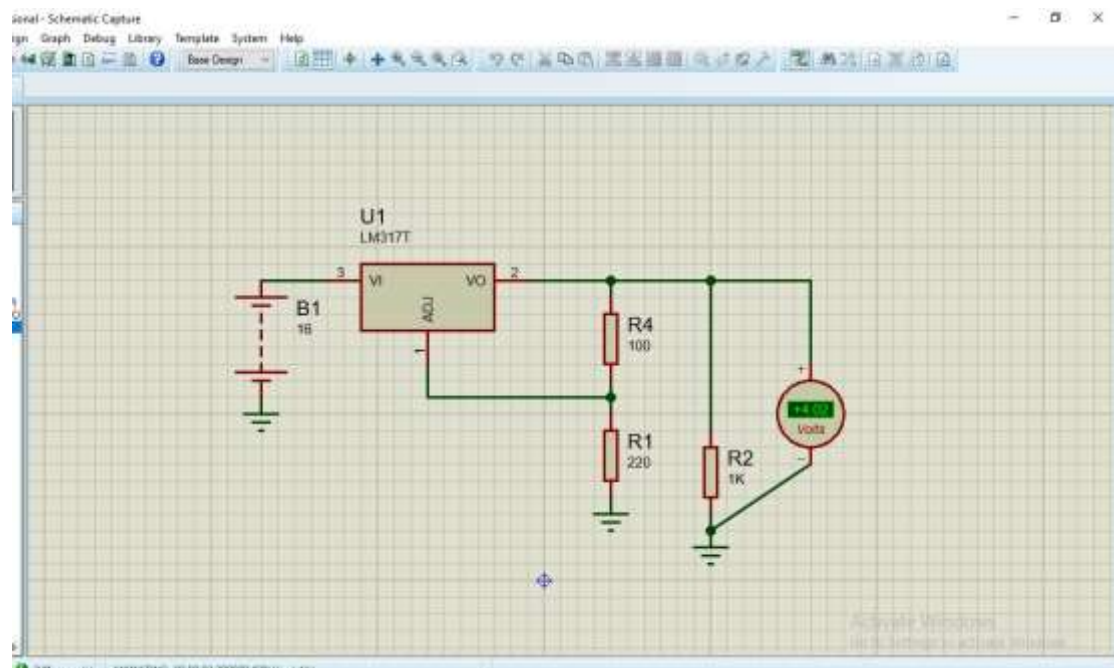
## The LM317 Adjustable Voltage Regulator (by Proteus)

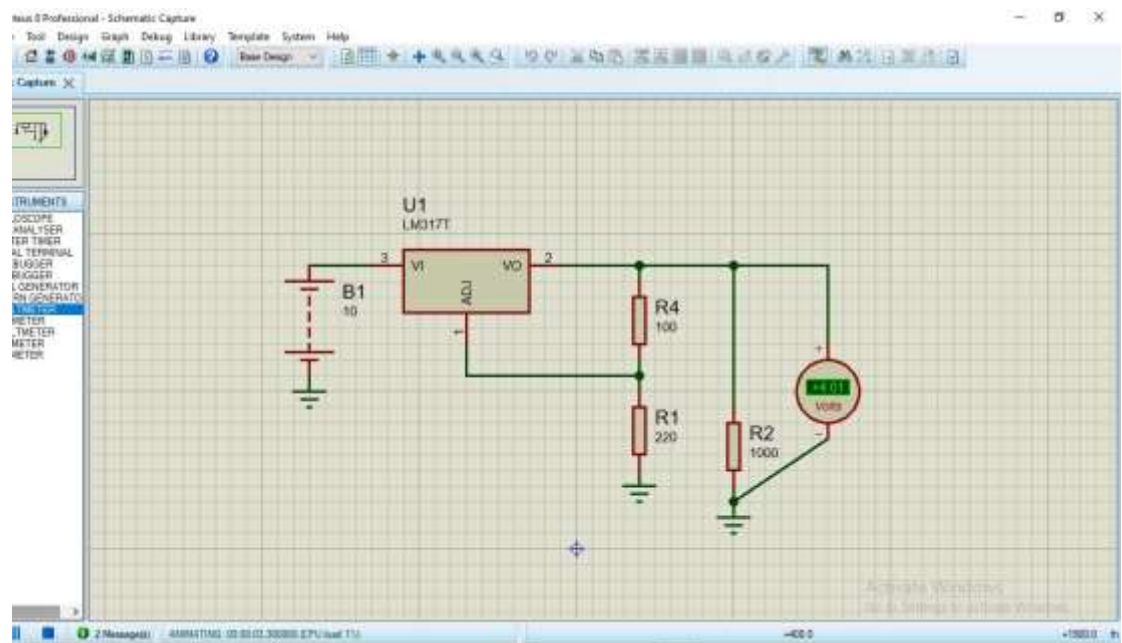
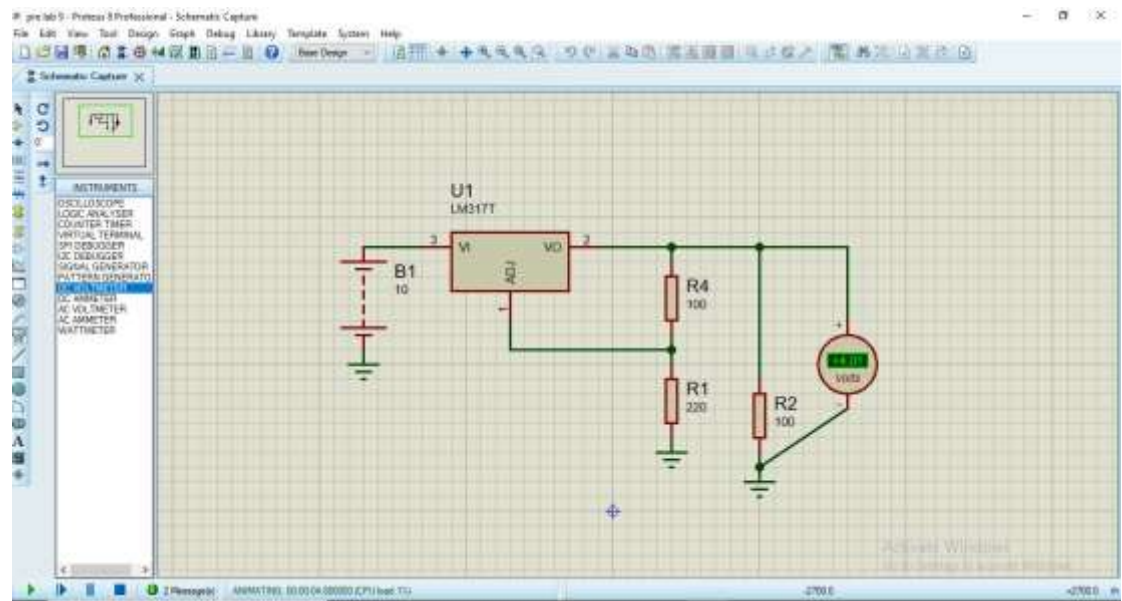


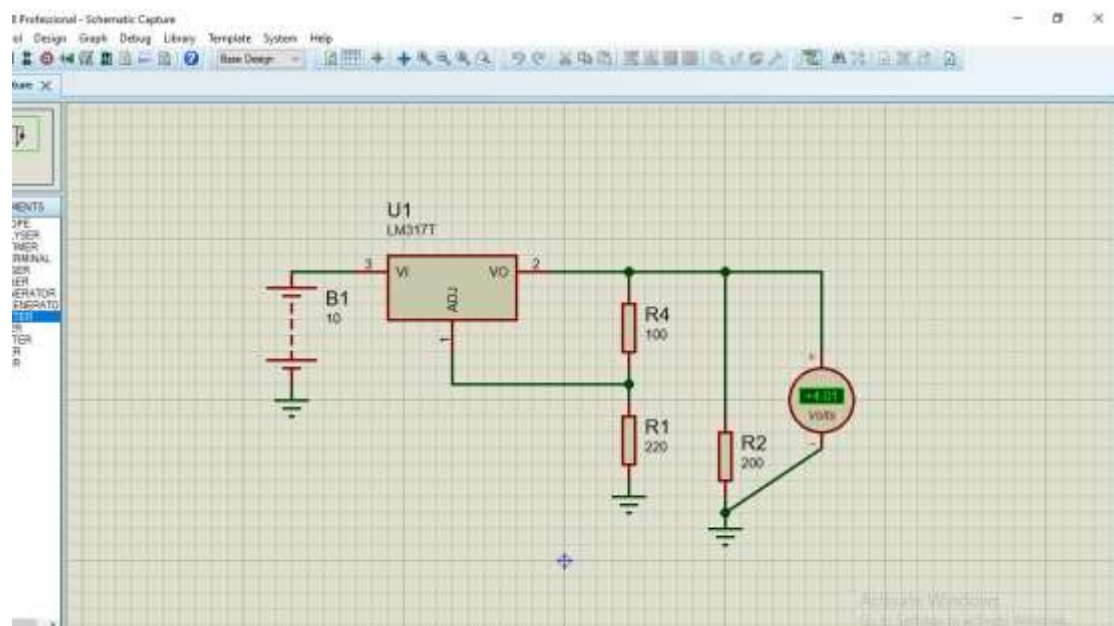
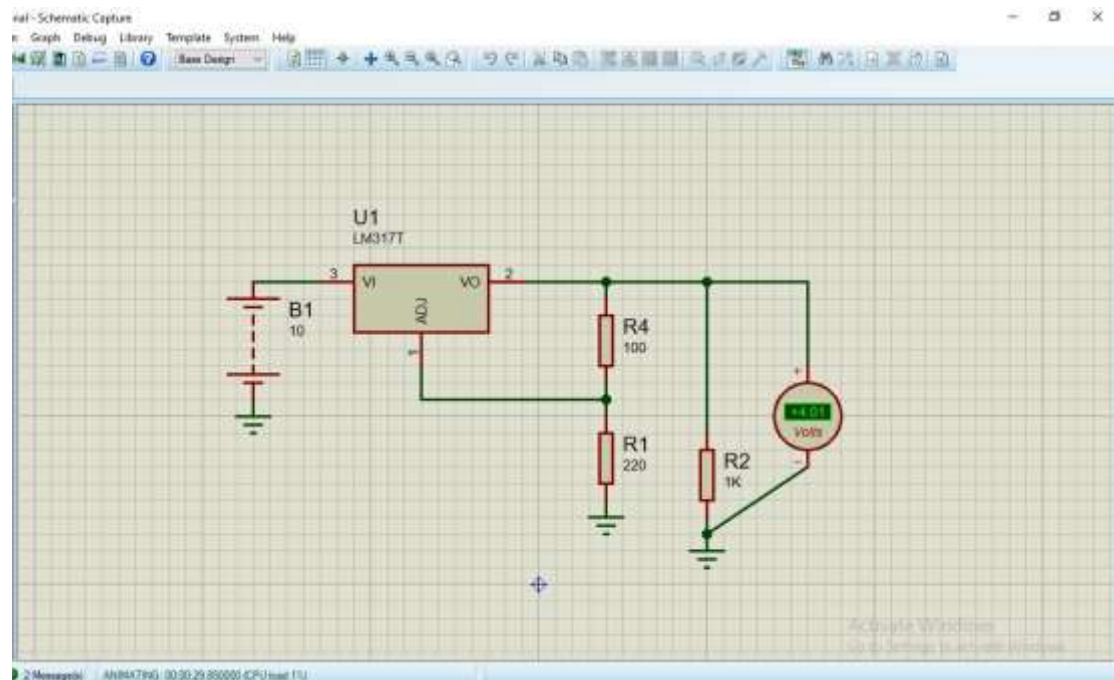












## Three Terminal Fixed Voltage Regulator 7805

